

Incidence Geometry of Projected Operator Composition

Rank Protection, Image–Kernel Alignment, and Promotion Limits

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Abstract

Problem. Let $A : \mathbb{C}^n \rightarrow \mathbb{C}^m$ and $B : \mathbb{C}^p \rightarrow \mathbb{C}^n$ be nonzero. Their product can nevertheless vanish. In a sectorized operator setting, this is the basic obstruction to promoting a Boolean support path through an intermediate sector to a nonzero projected composition.

Approach. We study the closed zero-product locus

$$Z = \{(A, B) : AB = 0\}$$

and its constructible nonzero-factor part. The identity $AB = 0 \iff \text{im } B \subseteq \ker A$ turns product failure into an image–kernel incidence condition. We stratify this condition by rank $A = r$ and by the double rank $(\text{rank } A, \text{rank } B) = (r, s)$.

Results. The fixed- r stratum has

$$\dim I_r = r(m + n - r) + p(n - r), \quad \text{codim } I_r = (m - r)(n - r) + pr.$$

The fixed- (r, s) incidence stratum has

$$\dim I_{r,s} = r(m + n - r) + s(n - r + p - s),$$

and has relative codimension rs inside the ambient rank- (r, s) matrix-pair stratum. Full column rank of A , or full row rank of B , excludes nonzero zero-products and gives quantitative singular-value lower bounds. A corrected computational audit distinguishes both-protected, left-only, right-only, unprotected-nonzero, and unprotected-zero routed products. In the declared Rubik realization, 2592 nonzero-factor routed records split into 2016 unprotected nonzero products and 576 machine-zero image–kernel alignments.

Boundary. These results concern one routed product. They do not identify a support path with a routed product, a routed product with a full word, a full word with a commutator, or low-order commutator support with Lie depth. Likewise, high codimension in the free matrix-pair space does not determine the pullback incidence geometry of a representation-derived family.

1 Introduction

The local question is elementary:

Why can $A \neq 0$ and $B \neq 0$ coexist with $AB = 0$?

The answer is exact:

$$\text{im } B \subseteq \ker A.$$

This condition is invisible to Boolean factor support. It depends on the relative position of two subspaces in the intermediate vector space. For projected operator blocks, it is therefore the first geometric datum that must be checked after a support path has been found.

The paper has three goals.

1. Give the rank-stratified geometry of the zero-product condition.
2. State the correct rectangular rank-protection criteria.
3. Separate local incidence from every stronger promotion involving route sums, words, commutators, Lie depth, or representation-derived parameter families.

The analysis is static: the sector projectors and operator family are fixed. Moving projectors, normal spectral charts, and parameter-dependent pullbacks remain research directions.

The image-kernel obstruction is adjacent to two independent questions. First, Boolean graph paths need not survive projected matrix composition. Second, nonzero word terms need not survive antisymmetrization into a commutator. These neighboring failures motivate the promotion table below, but the corresponding theorems are not assumed here. The incidence and rank-protection criteria can serve as route-survival gates, but do not promote routed products to word or Lie accessibility.

Notation and Claim Layers

The notation is introduced below in the order required by the routed-product problem. The claim-status vocabulary is **Theorem** for exact statements under their stated hypotheses, **Computational Certificate** only when a finite numerical check is explicitly assigned that status, **Computational Observation** for the declared Rubik, random-control, perturbation, and finite-atlas records, and **Research Program** for represented pullback geometry and all stronger promotion statements that remain open. The complete status table appears in Section 7.

Let V be a finite-dimensional complex Hilbert space with a complete orthogonal sectorization

$$V = \bigoplus_{i \in \mathcal{I}} E_i, \quad I = \sum_{i \in \mathcal{I}} Q_i, \quad Q_i Q_j = \delta_{ij} Q_i.$$

Let $\mathcal{Y} = \{Y_a\}_{a \in \mathcal{A}}$ be a declared finite operator family. No commutant assumption is imposed. For a sector triple (i, k, j) and ordered operator pair (a, b) , define

$$A_{ik}^a = Q_i Y_a Q_k : E_k \rightarrow E_i, \quad B_{kj}^b = Q_k Y_b Q_j : E_j \rightarrow E_k.$$

Write

$$d_i = \dim E_i, \quad d_k = \dim E_k, \quad d_j = \dim E_j.$$

The routed projected composition is

$$T_{ikj}^{a,b} = Q_i Y_a Q_k Y_b Q_j = A_{ik}^a B_{kj}^b.$$

Our arrow convention is

$$j \longrightarrow k \longrightarrow i.$$

A support path records only

$$A_{ik}^a \neq 0, \quad B_{kj}^b \neq 0.$$

It does not record whether $T_{ikj}^{a,b}$ is nonzero.
The full two-letter word block is a different object:

$$W_{ij}^{a,b} = Q_i Y_a Y_b Q_j = \sum_{k \in \mathcal{J}} T_{ikj}^{a,b}.$$

If the declared family is a skew-Hermitian Lie family $\mathcal{X} = \{X_g\}$, then the projected commutator is

$$C_{ij}^{g,h} = Q_i [X_g, X_h] Q_j = W_{ij}^{g,h} - W_{ij}^{h,g}.$$

These three objects – routed term, full word, and commutator – remain separate throughout the paper.

2 Image–Kernel Incidence

Theorem 2.1 (Image–Kernel Criterion).

Let

$$A : \mathbb{C}^n \rightarrow \mathbb{C}^m, \quad B : \mathbb{C}^p \rightarrow \mathbb{C}^n.$$

Then

$$AB = 0 \iff \text{im } B \subseteq \ker A.$$

Proof. *If $AB = 0$, then $A(Bv) = 0$ for every $v \in \mathbb{C}^p$, so every vector in $\text{im } B$ lies in $\ker A$. Conversely, if $\text{im } B \subseteq \ker A$, then $A(Bv) = 0$ for every v , hence $AB = 0$.*

2.1 Closed and constructible loci

Define

$$Z_{m,n,p} = \{(A, B) \in \text{Mat}_{m \times n}(\mathbb{C}) \times \text{Mat}_{n \times p}(\mathbb{C}) : AB = 0\}.$$

The entries of AB are polynomial in the entries of A and B , so $Z_{m,n,p}$ is a closed affine algebraic set.

The nonzero-factor incidence locus is

$$Z_{m,n,p}^\times = Z_{m,n,p} \cap \{A \neq 0\} \cap \{B \neq 0\}.$$

It is constructible, not generally a closed subvariety. Fixed exact-rank pieces are locally closed strata.

Proposition 2.2 (Ambient Generic Nonincidence).

The complement of $Z_{m,n,p}$ is a nonempty Zariski-open dense subset of the free matrix-pair space. Consequently, $Z_{m,n,p}$ and $Z_{m,n,p}^\times$ have Lebesgue measure zero.

Proof. *The zero-product equations are polynomial, and they do not vanish identically on the ambient matrix-pair space. Thus their common zero set is a proper closed algebraic subset. Its complement is nonempty, Zariski open, and dense; every proper complex algebraic subset has real Lebesgue measure zero.*

This proposition concerns freely varying matrix pairs. It makes no statement about a constrained parameter family whose image may lie partly or entirely inside $Z_{m,n,p}$.

3 Rank-Stratified Geometry

Theorem 3.1 (Fixed-A-Rank Incidence Stratum).

For $0 \leq r \leq \min(m, n)$, let

$$I_r = \{(A, B) : \text{rank } A = r, AB = 0\}.$$

Then I_r is locally closed and

$$\dim I_r = r(m + n - r) + p(n - r).$$

Its codimension in $\text{Mat}_{m \times n} \times \text{Mat}_{n \times p}$ is

$$\text{codim } I_r = (m - r)(n - r) + pr.$$

Proof. The rank- r matrices A form a locally closed stratum of dimension $r(m + n - r)$. For fixed A , the condition $AB = 0$ requires every column of B to lie in the $(n - r)$ -dimensional space $\ker A$. The fiber has dimension $p(n - r)$. Adding base and fiber dimensions gives the first formula. Subtracting from the ambient dimension $mn + np$ gives the codimension formula.

For the nonzero-factor locus, the admissible A -ranks are exactly

$$1 \leq r \leq \min(m, n - 1).$$

The upper bound is $r < n$, not $r < \min(m, n)$. In particular, if $m < n$, a full-row-rank matrix with $r = m$ still has a nontrivial kernel and can participate in a nonzero zero-product.

Theorem 3.2 (Fixed Double-Rank Incidence Stratum).

Let

$$I_{r,s} = \{(A, B) : \text{rank } A = r, \quad \text{rank } B = s, \quad AB = 0\}.$$

This stratum is nonempty exactly when

$$0 \leq r \leq \min(m, n), \quad 0 \leq s \leq \min(n - r, p).$$

When nonempty,

$$\dim I_{r,s} = r(m + n - r) + s(n - r + p - s).$$

Proof. Choose a rank- r matrix A , contributing $r(m + n - r)$ dimensions. Its kernel has dimension $n - r$. A rank- s map $B : \mathbb{C}^p \rightarrow \ker A$ exists exactly when $s \leq \min(n - r, p)$ and belongs to a rank- s matrix stratum of dimension $s(n - r + p - s)$.

Corollary 3.3 (Relative Codimension by Rank Product).

Inside the ambient rank-pair stratum

$$\{(A, B) : \text{rank } A = r, \text{ rank } B = s\},$$

the incidence condition $AB = 0$ has relative codimension

$$rs.$$

Proof. The ambient rank-pair stratum has dimension

$$r(m + n - r) + s(n + p - s).$$

Subtracting the dimension in Theorem 3.2 gives rs .

This formula isolates the alignment cost after the two ranks have already been fixed. Rank deficiency and image-kernel alignment are distinct conditions.

Corollary 3.4 (Square-Block Asymptotics).

For $m = n = p = d$ and $1 \leq r \leq d - 1$,

$$\text{codim } I_r = (d - r)^2 + dr = d^2 - dr + r^2.$$

The minimum is attained at the integer or integers nearest $d/2$, and equals

$$\left\lceil \frac{3d^2}{4} \right\rceil.$$

Thus the dominant nonzero-factor incidence stratum has codimension asymptotic to $3d^2/4$ in the $2d^2$ -dimensional free matrix-pair space.

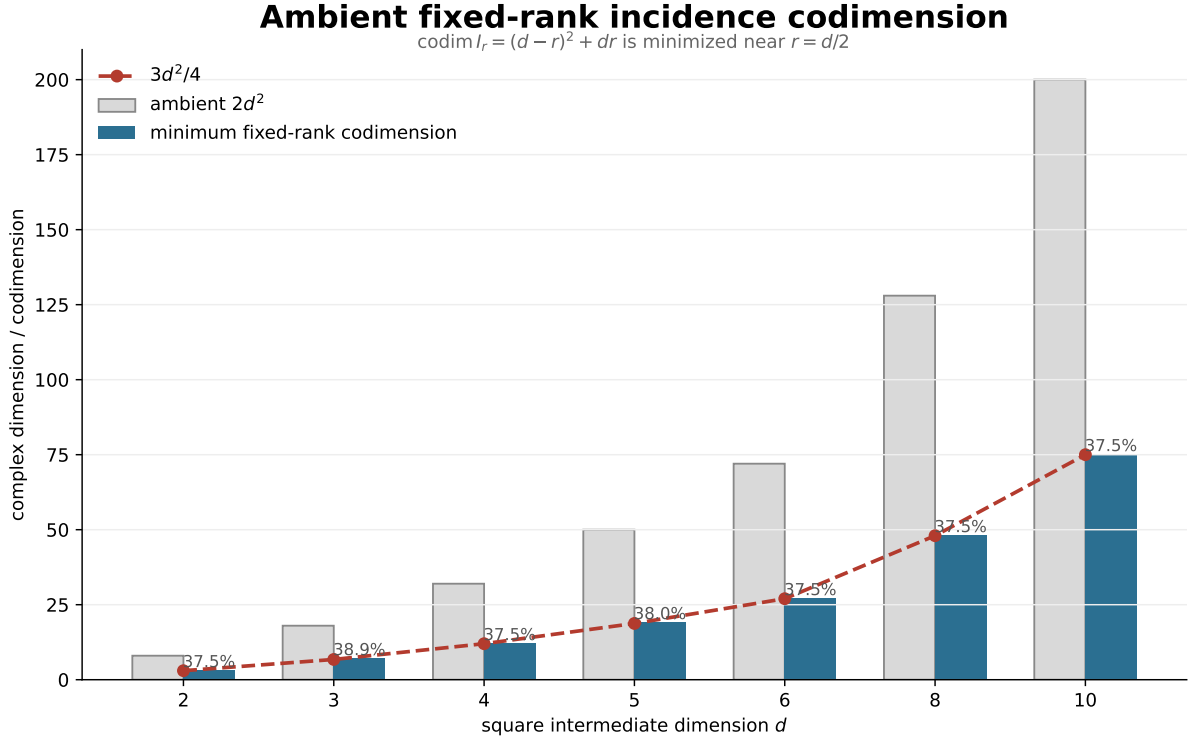


Figure 1. Exact fixed-rank codimension values for square blocks. The blue bars show the minimum of $\text{codim } I_r = (d - r)^2 + dr$ over admissible nonzero ranks, against the $2d^2$ ambient dimension and the $3d^2/4$ asymptotic. This is an ambient free-matrix calculation, not a represented pullback codimension.

4 Rank Protection

Return to a sector triple:

$$A : E_k \rightarrow E_i, \quad B : E_j \rightarrow E_k, \quad d_k = \dim E_k.$$

Define

$$L = [\text{rank } A = d_k], \quad R = [\text{rank } B = d_k].$$

The condition L means that A has full column rank, which is possible only when $d_i \geq d_k$. Similarly, the condition R means that B has full row rank, which is possible only when $d_j \geq d_k$.

Theorem 4.1 (Rank-Protected Product Survival).

Let $A \neq 0$ and $B \neq 0$.

1. If $\text{rank } A = d_k$, then $AB \neq 0$.
2. If $\text{rank } B = d_k$, then $AB \neq 0$.

More quantitatively,

$$\|AB\|_F \geq \sigma_{\min}(A)\|B\|_F$$

when A has full column rank, and

$$\|AB\|_F \geq \sigma_{\min}(B)\|A\|_F$$

when B has full row rank.

Proof. Full column rank gives $\ker A = 0$, so Theorem 2.1 rules out $AB = 0$ for nonzero B . Applying the smallest-singular-value inequality to each column of B gives the first norm bound. For the dual statement, use $(AB)^* = B^*A^*$ and note that B^* has full column rank with the same positive singular values as B .

The condition

$$\text{rank } A = \min(d_i, d_k)$$

is not sufficient for left protection. If $d_i < d_k$, such a matrix has maximum possible rank but still has a nonzero kernel. The analogous warning holds for B when $d_j < d_k$.

The five mutually exclusive audit classes are:

Class	Condition
both-protected	$L \wedge R$
left-only	$L \wedge \neg R$
right-only	$\neg L \wedge R$
unprotected-nonzero	$\neg L \wedge \neg R \wedge AB \neq 0$
unprotected-zero	$\neg L \wedge \neg R \wedge AB = 0$

Protected records are still required to evaluate AB numerically. Rank protection is used as a theorem-level cross-check, not as a branch that skips the product calculation.

5 Promotion Limits

The local incidence theorem resolves only the first matrix-composition gate. For fixed i, k, j, g, h , the following implications are invalid without additional hypotheses:

Local datum	Stronger conclusion	Missing promotion condition
$A \neq 0, B \neq 0$	$T_{ikj}^{g,h} \neq 0$	image-kernel nonalignment or rank protection
some $T_{ikj}^{g,h} \neq 0$	$W_{ij}^{g,h} \neq 0$	no cancellation in the sum over k
$W_{ij}^{g,h} \neq 0$ and $W_{ij}^{h,g} \neq 0$	$C_{ij}^{g,h} \neq 0$	no antisymmetric cancellation
low-order Lie support	finite or exact $D_{\text{Lie}}(i, j)$	higher Hall data and closure/saturation certificate
high ambient codimension	rare represented incidence	nondegenerate pullback or transversality of the parameter map

Equivalently,

$$\begin{aligned} \text{support path} &\not\Rightarrow T_{ikj}^{g,h} \neq 0, \\ T_{ikj}^{g,h} \neq 0 &\not\Rightarrow W_{ij}^{g,h} \neq 0, \\ W_{ij}^{g,h}, W_{ij}^{h,g} \neq 0 &\not\Rightarrow C_{ij}^{g,h} \neq 0. \end{aligned}$$

The first failure is controlled exactly by image–kernel incidence for the declared route. The second is a sum-over-routes cancellation problem. The third is an antisymmetrization problem. None is interchangeable with the others.

5.1 Ambient versus represented incidence

Let a constrained family be parameterized by

$$\Phi : \Theta \longrightarrow \text{Mat}_{m \times n} \times \text{Mat}_{n \times p}.$$

Its incidence locus is

$$\Phi^{-1}(Z_{m,n,p}).$$

Ambient codimension does not determine the dimension of this pullback. A promotion from ambient genericity to a represented family requires, at minimum, evidence that $\Phi(\Theta)$ is not contained in $Z_{m,n,p}$ and a suitable transversality or nondegeneracy statement on the relevant rank strata. Symmetry may force a family into incidence despite the high ambient codimension.

6 Exact and Computational Case Studies

The displayed integer examples are exact. All remaining claims in this section are finite numerical observations and are not used to prove the incidence theorems.

6.1 Declared numerical policy

The projected-composition audit uses:

Quantity	Policy
support	$\ A\ _F, \ B\ _F > 10^{-8}$
numerical rank	singular values above $\max(10^{-12}, 10^{-9}\sigma_{\max})$
product nonzero	$\ AB\ _F > 10^{-12} + 10^{-10}\ A\ _F\ B\ _F$
normalized product	$\eta(A, B) = \ AB\ _F / (\ A\ _F\ B\ _F)$
image–kernel action	$\ AU_B\ _F$, where U_B spans $\text{im } B$
subspace residual	$\delta_{\text{inc}}(A, B) = \ (I - P_{\ker A})U_B\ _F$

Here the columns of U_B are an orthonormal basis of $\text{im } B$ and $P_{\ker A}$ is the orthogonal projector onto $\ker A$. The reported δ_{inc} is not divided by $\sqrt{\text{rank } B}$. Small $\eta(A, B)$ is a numerical alignment diagnostic, not an exact incidence proof. Every protected record is still multiplied, and a numerically protected zero-product triggers an assertion.

6.2 Exact integer examples

Five explicit integer matrix pairs realize all five audit classes. In the table, $r_A = \text{rank } A$ and $r_B = \text{rank } B$.

Example	(m, n, p)	(r_A, r_B)	Outcome
both	$(2, 2, 2)$	$(2, 2)$	$AB \neq 0$
left only	$(3, 2, 1)$	$(2, 1)$	$AB \neq 0$
right only	$(1, 2, 3)$	$(1, 2)$	$AB \neq 0$
unprotected, nonzero	$(2, 2, 2)$	$(1, 1)$	$AB \neq 0$
unprotected, zero	$(2, 2, 2)$	$(1, 1)$	$AB = 0$

A concrete integer realization is

1. both protected: $A = I_2$ and $B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$;
2. left only: $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$;
3. right only: $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$;
4. unprotected and nonzero: $A = B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$;
5. unprotected and zero: $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$.

Their ranks and products can be checked directly. In particular, the last two examples demonstrate that a lack of protection is necessary but not sufficient for incidence.

6.3 Rubik routed-product registration

The computational Rubik case study fixes:

- the nine declared center-decomposition sectors, whose dimensions are

$$(20, 2, 39, 26, 1, 39, 66, 8, 27);$$

- the 18 skew-Hermitian operators $X_g = (\rho(g) - \rho(g)^*)/2$;
- all ordered operator pairs, including repeated operators;
- all sector triples with off-diagonal factor legs $i \neq k$ and $k \neq j$.

For each order-two half turn, unitarity gives $\rho(g)^* = \rho(g)^{-1} = \rho(g)$, and therefore $X_g = 0$. The audit confirms six zero operators, at zero-based indices 2, 5, 8, 11, 14, 17, with maximum norm exactly zero in the declared realization. Consequently, all nonzero-factor routed records arise from the twelve quarter-turn operators. This explains the count $144 = 12^2$ for each exceptional sector triple, even though the declared family contains 18 operators and the enumeration initially considers all ordered pairs.

This anti-Hermitian-part registration is not the numerical matrix-logarithm family used in Paper V's S_4 case study [2]. In particular, a half turn vanishes here but generally has a nonzero logarithmic generator. The two papers' R_1^{Lie} and R_2^{Lie} censuses therefore cannot be compared without an explicit observable-family alignment.

The corrected census is:

Class	Count
both-protected	0
left-only	0
right-only	0
unprotected-nonzero	2016
unprotected-zero	576
total nonzero-factor routes	2592

The 576 machine-zero products are concentrated in four zero-based sector triples:

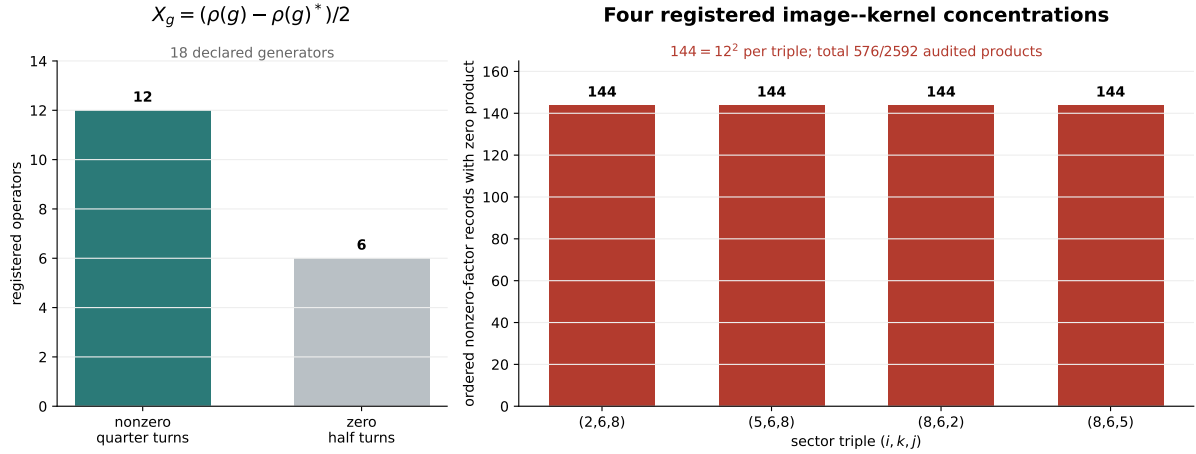
(i, k, j)	Count
(2, 6, 8)	144
(5, 6, 8)	144
(8, 6, 2)	144
(8, 6, 5)	144

Across these records,

$$\max \|AB\|_F = 2.467 \times 10^{-16}, \quad \max \eta(A, B) = 6.737 \times 10^{-17}.$$

The minimum factor norm is 1.803. The maximum image–kernel action residual is 1.210×10^{-15} and the maximum subspace distance residual is 9.485×10^{-15} . These are machine-zero image–kernel alignment records in the declared complex128 realization, not exact symbolic equalities.

Declared Rubik routed-product incidence census



This is a finite numerical registration; it does not identify a represented pullback codimension.

Figure 2. The declared Rubik routed-product census. Six half-turn operators vanish under the anti-Hermitian-part registration, leaving twelve nonzero quarter-turn operators and therefore $144 = 12^2$ nonzero-factor records in each of four registered image–kernel concentrations. The display is a finite numerical registration and does not determine the codimension of a representation-derived pullback locus.

6.4 Random and rank-deficient controls

Five dense random systems contribute 1620 routed records. Every record is both-protected and has nonzero product.

Five controls with an 80% rank-deficient block-generation probability contribute another 1620 records:

Class	Count
both-protected	122
left-only	274
right-only	274
unprotected-nonzero	950
unprotected-zero	0

Thus, rank deficiency creates unprotected routes but does not, by itself, force image–kernel incidence.

For independent dense square matrix pairs, the normalized product audit gives:

d	$\min \eta$	$q_{0.001}$	$q_{0.01}$	median
2	0.1068	0.2322	0.3461	0.7070
3	0.2235	0.3017	0.3678	0.5744
4	0.2460	0.3144	0.3610	0.4975
5	0.2444	0.3122	0.3463	0.4455

Each dimension uses 100000 samples. No sample falls below relative thresholds 10^{-8} , 10^{-10} , or 10^{-12} . This is an implementation and threshold sanity check consistent with Proposition 2.2; it is not evidence for operator or Lie completion.

6.5 Perturbation and finite full-array atlas

Constructed incidence pairs in dimensions 2, 3, 4 were perturbed at $\varepsilon = 10^{-6}, 10^{-4}, 10^{-2}$. All 900/900 perturbed products became nonzero at the declared threshold.

A separate finite Lie audit registers 80 eight-dimensional skew-Hermitian systems in four mask families. It stores complete generator-indexed R_1^{Lie} arrays, complete commutator-indexed R_2^{Lie} arrays, complete D_{Lie} matrices, each per-depth support array, each cumulative support array, and deterministic SHA-256 hashes.

Each of the four threshold-defined low-order array classes contains 20 systems. Within every class:

- the complete D_{Lie} arrays agree elementwise;
- all per-depth and cumulative support arrays agree elementwise;
- the results are stable at thresholds $10^{-9}, 10^{-8}, 10^{-7}$;
- the cumulative generator–commutator basis passes the closure audit described below.

At rank tolerance 10^{-8} , the cumulative dimensions by mask family, listed through the first empty round, are

1. family 0: (3, 6, 14, 28, 49, 63, 63), first empty round 6;
2. family 1: (3, 5, 10, 20, 40, 59, 63, 63), first empty round 7;
3. family 2: (3, 5, 10, 18, 31, 43, 51, 58, 61, 63, 63), first empty round 10;
4. family 3: (3, 6, 14, 32, 63, 63), first empty round 5.

An empty round means that every commutator of a basis element in the newest layer with every declared generator produces no new vector after projection against the cumulative basis. The closure audit additionally checks all generator–basis pairs, not solely those involving the newest layer. If $\mathcal{L} = \text{span}\{L_a\}$ is the final numerical basis, it forms

$$R_{\text{cl}} = [(I - P_{\mathcal{L}}) \text{vec}[X_g, L_a]]_{g,a}.$$

Across all 80 systems, the final dimension is 63, the smallest retained singular value of the vectorized basis matrix is at least 0.999999999998, the largest singular value of R_{cl} is at most 7.175×10^{-13} , and the largest individual closure residual is 6.627×10^{-13} . Thus all generator–basis commutators lie in the final span to well below the declared rank tolerance. This is a numerical closure certificate for the finite atlas, not an exact-arithmetic Lie-algebra theorem.

No disagreement occurs in this finite atlas. This is an observation about the declared four families, not a theorem that $(R_1^{\text{Lie}}, R_2^{\text{Lie}})$ determines D_{Lie} .

7 Claim Status and Boundary

The table uses the four claim levels. Corollaries and exact finite examples belong to the Theorem level; refuted or unclaimed promotions are listed separately as boundaries.

Claim	Status
Image–Kernel Criterion	Theorem
$Z = \{AB = 0\}$ is closed; $Z^\times$ is constructible	Theorem
fixed- r dimension and codimension	Theorem
fixed- (r, s) dimension and relative codimension rs	Theorem
square-block dominant codimension	Theorem
left/right rank protection and norm bounds	Theorem
existence of the five displayed exact integer examples	Theorem
six half-turn zeros and corrected Rubik routed census	Computational Observation
random quantiles and perturbation breakdown	Computational Observation
full-array finite Lie atlas agreement and closure residual	Computational Observation
low-order Lie support determines Lie depth	Research Program
represented pullback incidence and transversality	Research Program

The boundary is explicit: graph-to-route, route-to-word, and word-to-commutator promotions are not claimed. Furthermore, ambient codimension does not determine represented pullback geometry without additional hypotheses.

There is no theorem or conjecture in this paper asserting

$$(R_1^{\text{Lie}}, R_2^{\text{Lie}}) \implies D_{\text{Lie}}.$$

8 Research Program

8.1 Conditional Low-Order Promotion Problem

A future promotion theorem would need explicit, independently checkable hypotheses rather than an undefined richness condition. Candidate gates include:

Gate	Candidate requirement
H1 low-order coverage	every finite-depth target already appears at the declared R_1^{Lie} or R_2^{Lie} layer
H2 route survival	every required routed product avoids image–kernel incidence
H3 cancellation control	route sums and antisymmetrizations do not erase the required channels, or are repaired in a declared Hall layer
H4 closure	the numerical or exact Lie closure is certified saturated

H1 is an exclusionary hypothesis, not an observed general principle. Indeed, it fails in the declared S4 numerical realization of Paper V: two registered channels have $R_1^{\text{Lie}} = R_2^{\text{Lie}} = 0$ but first appear at $D_{\text{Lie}} = 2$ [2]. The table is a research checklist, not a theorem statement.

8.2 Pullback incidence

For a representation-derived map $\Phi : \Theta \rightarrow \text{Mat} \times \text{Mat}$, determine:

1. whether $\Phi(\Theta)$ is contained in an incidence stratum;
2. whether Φ is transverse to the relevant fixed-rank strata;
3. how symmetry changes the pullback codimension;
4. whether ranks remain constant on the declared parameter chart.

The object of interest is the pullback locus $\Phi^{-1}(Z)$, not the ambient codimension alone.

8.3 Structural stability and moving charts

The reported perturbation audit varies free matrix entries. A structural stability theorem would instead perturb inside a representation-preserving or sector-preserving family. Moving projectors require normal spectral charts and coherent continuation before incidence can be compared across parameters.

8.4 Hub recurrence

The four Rubik zero-product triples share the intermediate sector $k = 6$. Whether such incidence concentration relates to transport centrality remains open. A formal relation would require aligned operator families and a theorem connecting graph, routed, and representation-derived data.

9 Conclusion

Projected factors compose precisely when the image of the right factor is not contained in the kernel of the left factor. Thus

$$AB = 0 \iff \text{im } B \subseteq \ker A$$

is the local geometric obstruction, while full column rank of A or full row rank of B protects a nonzero product. On a fixed double-rank stratum, the incidence condition has relative codimension rs , cleanly separating rank deficiency from the additional image–kernel alignment cost.

In the declared Rubik realization, the finite audit records 2016 unprotected nonzero products and 576 machine-zero image–kernel alignments among 2592 nonzero-factor routes. This is a Computational Observation, not an exact represented-incidence theorem.

These conclusions stop at routed matrix composition. They do not identify a support path with a routed product, a routed product with a full word, a word with a commutator, or low-order support with finite Lie depth. Rank protection, image–kernel alignment, and promotion limits are therefore distinct parts of one static composition problem.

10 Related Work and Novelty Boundary

The matrix inequalities and singular-value bounds are standard [7]. Determinantal and incidence strata are standard objects in algebraic geometry [6, 5, 4]. The distinction between Boolean zero patterns and evaluated matrix products is adjacent to qualitative matrix theory and structural controllability [1, 8]. Routed products may also be viewed through path and quiver evaluation, although this paper does not construct a path-algebra quotient [9].

Related work separates support graphs from projected composition and direct support from words, commutators, and low-depth Lie data [3, 2]. Those results are not used in the incidence proofs here.

Appendix A Computational Artifacts

The following repository artifacts support the exact incidence tables and the computational case studies. The default directory is `experiments/paper7/`; paths are relative to that directory.

Artifact	Role	Short path
A1	exact fixed-rank and fixed-double-rank tables	<code>validation/incidence_variety_codim.py</code>
A2	routed-product, rank-protection, and incidence audit	<code>validation/rank_protected_bridge_audit.py</code>
A3	full-array typed low-order and Lie-depth audit	<code>validation/atlas_r2_boundary.py</code>
A4	generated incidence tables	<code>results/incidence_geometry.json, .txt</code>
A5	generated projected-composition witnesses	<code>results/projected_composition_audit.json, .txt</code>
A6	generated complete-array Lie atlas	<code>results/full_array_lie_atlas.json, .txt</code>

From the repository root, run an executable artifact as `python experiments/paper7/<short path>`. A4 contains the fixed-rank tables; A5 records thresholds, exact examples, the Rubik operator-family declaration, zero-product witnesses, random controls, and perturbations; A6 records full arrays, per-layer and cumulative supports, dimensions, closure certificates, and hashes.

Every zero-product witness records

$$(d_i, d_k, d_j), \quad (\text{rank } A, \text{rank } B), \quad \sigma_{\min}^+(A), \quad \sigma_{\min}^+(B),$$

together with $\|A\|_F$, $\|B\|_F$, $\|AB\|_F$, $\eta(A, B)$, $\|AU_B\|_F$, the image-to-kernel distance, sector indices, and operator indices.

The array hashes detect artifact drift. Mathematical equality claims in the finite atlas use `numpy.array_equal` on the complete arrays rather than hash equality alone.

All listed artifacts are available in the [RIME repository](#).

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